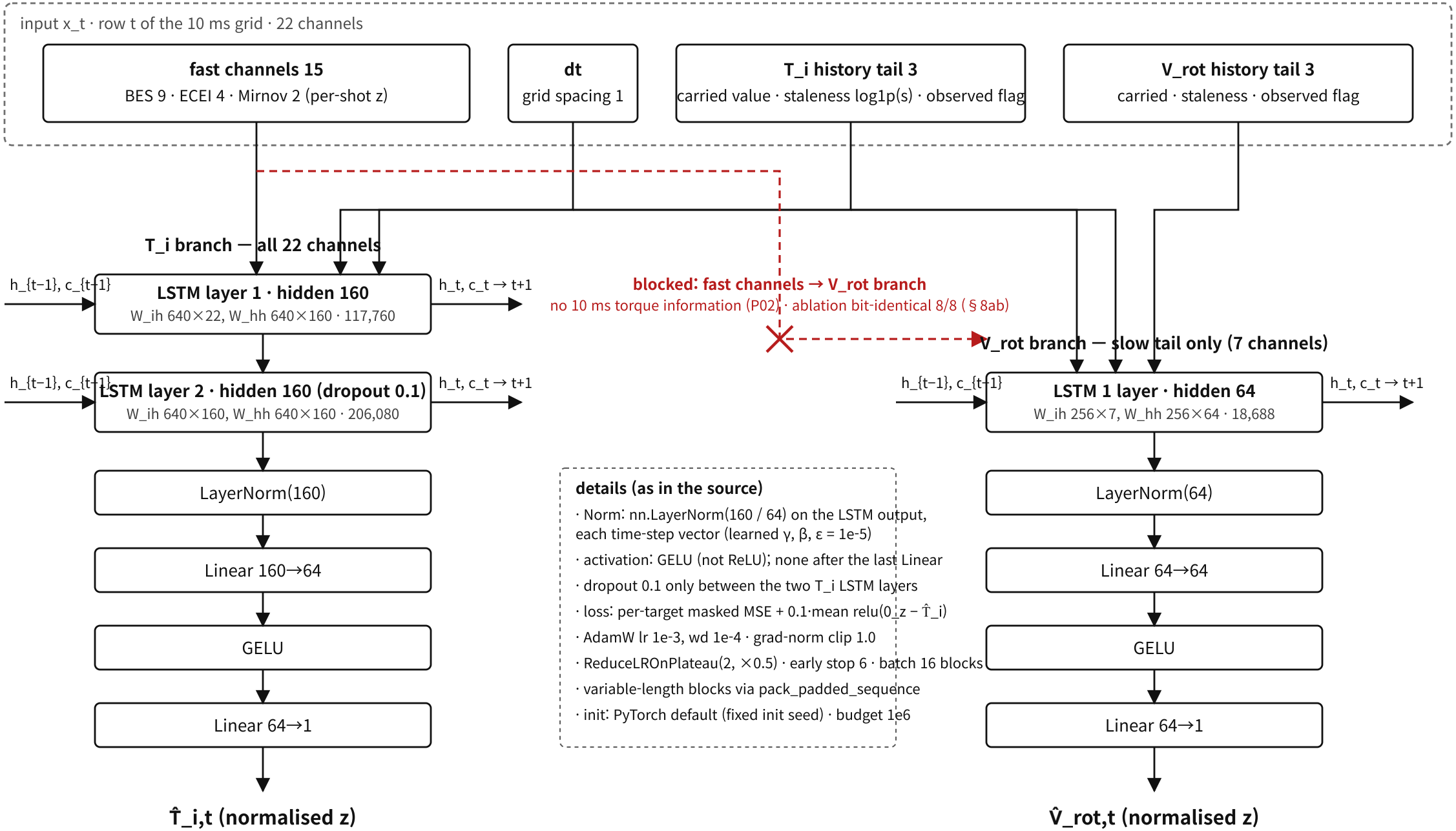


loss: per-target masked MSE on rows where CES was actually measured · the CES input at the target row is masked to 0 · 357,570 parameters (T_i branch 334,529 · V_rot branch 23,041)

State flows only from past to present (causal). One step per grid row; empty rows are inputs too (carried value, staleness and flag turn missingness into information).